

**US Geological Survey
Alaska Science Center
Statement of Work**

1.0 INTRODUCTION

The USGS needs to acquire apatite fission-track (AFT) and U-Pb analyses from geological samples collected from the Kuskokwim mountains (Km) region, Alaska.

2.0 BACKGROUND

The USGS conducts regional geologic and metallogenic framework studies that yield insight to the distribution of critical mineral resources in the United States. AFT thermochronometry is a technique of isotopic dating that can elucidate the timing, rates, and magnitude of rock exhumation. Combined AFT and apatite U-Pb analysis of bedrock samples from the Donlin mine and other prospects in the Km will be used to test hypotheses about the depth of formation and post-formation exhumation of the deposit--critical information in refining conceptual models of ore deposit formation. Samples are also distributed along the Iditarod-Nixon Fork fault system and are anticipated to reveal phases of deformation along this fault system, particularly how ore deposits that formed with or prior to faulting were translated. These constraints are important to both understanding the regional tectonic evolution of the area, as well as testing hypotheses regarding the spatiotemporal relation between fault activity and mineralization in the region. The second planned use of this data is to create regional exhumation maps to inform next-generation mineral prospectivity mapping efforts.

3.0 SCOPE

AFT+U-Pb thermochronometry requires access to and expertise in the use of all supplies and instrumentation for preparing samples and counting and characterizing spontaneous fission-tracks (e.g., sample mounting, track etching by immersion in HNO₃, and precision microscope capable of at least 2000x magnification). Critical to scientific objectives are measurements of track length that follow modern best practices (e.g., ²⁵²Cf irradiation), which additionally requires access to a vacuum chamber and fission fragments from a ²⁵²Cf source. Measurement of grain-by-grain Uranium and Calcium isotopes for final AFT age calculation and Uranium and Lead isotopes for complimentary U-Pb age calculation requires access to and expertise in a Nd: YAG (or similar) laser and sample cell linked to an inductively coupled plasma mass spectrometer (LA-ICP-MS)

USGS personnel will provide samples in two forms (1) vials of preconcentrated apatite, and (2) bulk rock samples. Bulk rocks samples will require mineral separation prior to AFT analysis to also be carried out by the vendor. The total work order is for 20 mineral separates and 25 total AFT+U-Pb analyses.

4.0 APPLICABLE DOCUMENTS

Publications documenting state-of-the-art analytical methods for collection and interpretation of AFT+U-Pb data:

- Donelick, R. A., & Miller, D. S. (1991). Enhanced TINT fission track densities in low spontaneous track density apatites using ^{252}Cf -derived fission fragment tracks: A model and experimental observations. *International Journal of Radiation Applications and Instrumentation. Part D. Nuclear Tracks and Radiation Measurements*, 18(3), 301-307.
- Donelick, R. A., O'Sullivan, P. B., & Ketcham, R. A. (2005). Apatite fission-track analysis. *Reviews in Mineralogy and Geochemistry*, 58(1), 49-94.
- Gleadow, A. J., Belton, D. X., Kohn, B. P., & Brown, R. W. (2002). Fission track dating of phosphate minerals and the thermochronology of apatite. *Reviews in mineralogy and geochemistry*, 48(1), 579-630.
- Hasebe, N., Barbarand, J., Jarvis, K., Carter, A., & Hurford, A. J. (2004). Apatite fission-track chronometry using laser ablation ICP-MS. *Chemical Geology*, 207(3-4), 135-145.
- Ketcham, R. A., Carter, A., Donelick, R. A., Barbarand, J., & Hurford, A. J. (2007). Improved modeling of fission-track annealing in apatite. *American Mineralogist*, 92(5-6), 799-810.
- Ketcham, R. A., Carter, A., Donelick, R. A., Barbarand, J., & Hurford, A. J. (2007). Improved measurement of fission-track annealing in apatite using c-axis projection. *American Mineralogist*, 92(5-6), 789-798.

5.0 TECHNICAL REQUIREMENTS

The contractor shall have experience in all aspects of AFT+U-Pb thermochronometry using conventional track counting techniques, established calibrated against community standards, ^{252}Cf irradiation, and LA-ICP-MS methods.

6.0 DELIVERABLES

The contractor shall provide deliverables in a report in Excel spreadsheet format that includes: Raw spontaneous fission-track counts and count area; associated kinetic parameters (e.g., Dpar and/or rmr0); raw and analytical standard-corrected measurements of ^{40}Ca , ^{147}Sm , ^{204}Pb , ^{206}Pb , ^{207}Pb , ^{232}Th , and ^{238}U by LA-ICP-MS; fission-track lengths, angles to grain c-axis, and associated kinetic parameters; analyst calibration factors (e.g., ζ and uncertainty); and final calculated AFT and U-Pb dates.

7.0 PERIOD OF PERFORMANCE

Award to December 31, 2026